Physics Worksheet

Magnetic Force

Name: _____ **Student #:** _____ **Student Email #:** _____

1. A coil is made of N loops of current-carrying wire packed closely together. What is the magnetic field at the centre of the coil?

2. What happens when two parallel wires are carrying charge in the same direction close to each other? Why?

3. Does Ampère's law prove that there is no magnetic field in a loop that encloses no current? Provide an example with a diagram where poor use of Ampère's law might appear to give this result

4. Derive the current density of a solenoid of radius R emitting a magnetic field B with N loops?

5. Derive the current density of a toroid of radius R emitting a magnetic field B with N loops?

6. What would happen to the current density and the magnetic field of a solenoid, if you were to make the two ends meet and create a toroid, such that the radius of the toroid is equal to the radius of one of the loops from the solenoid?